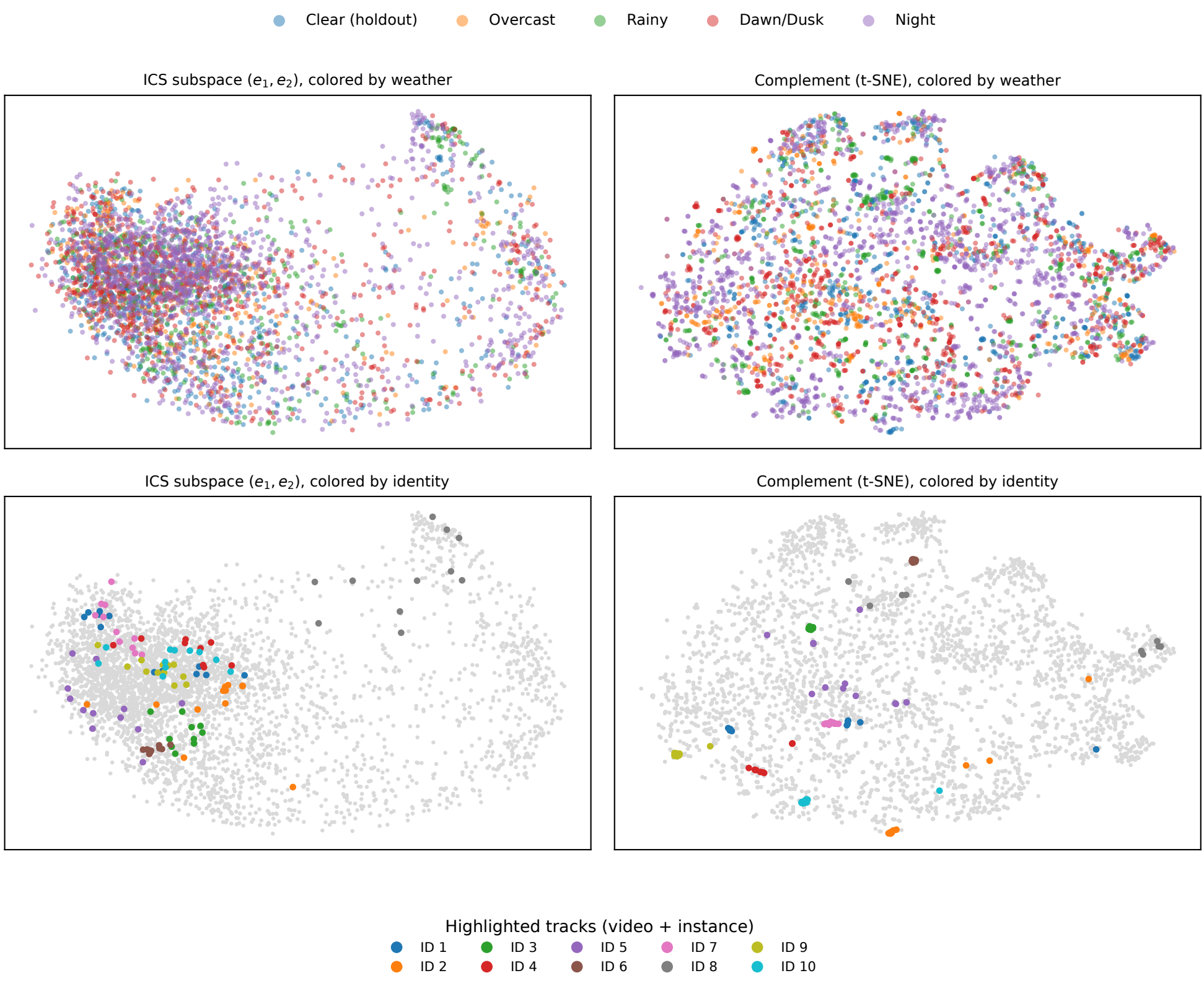
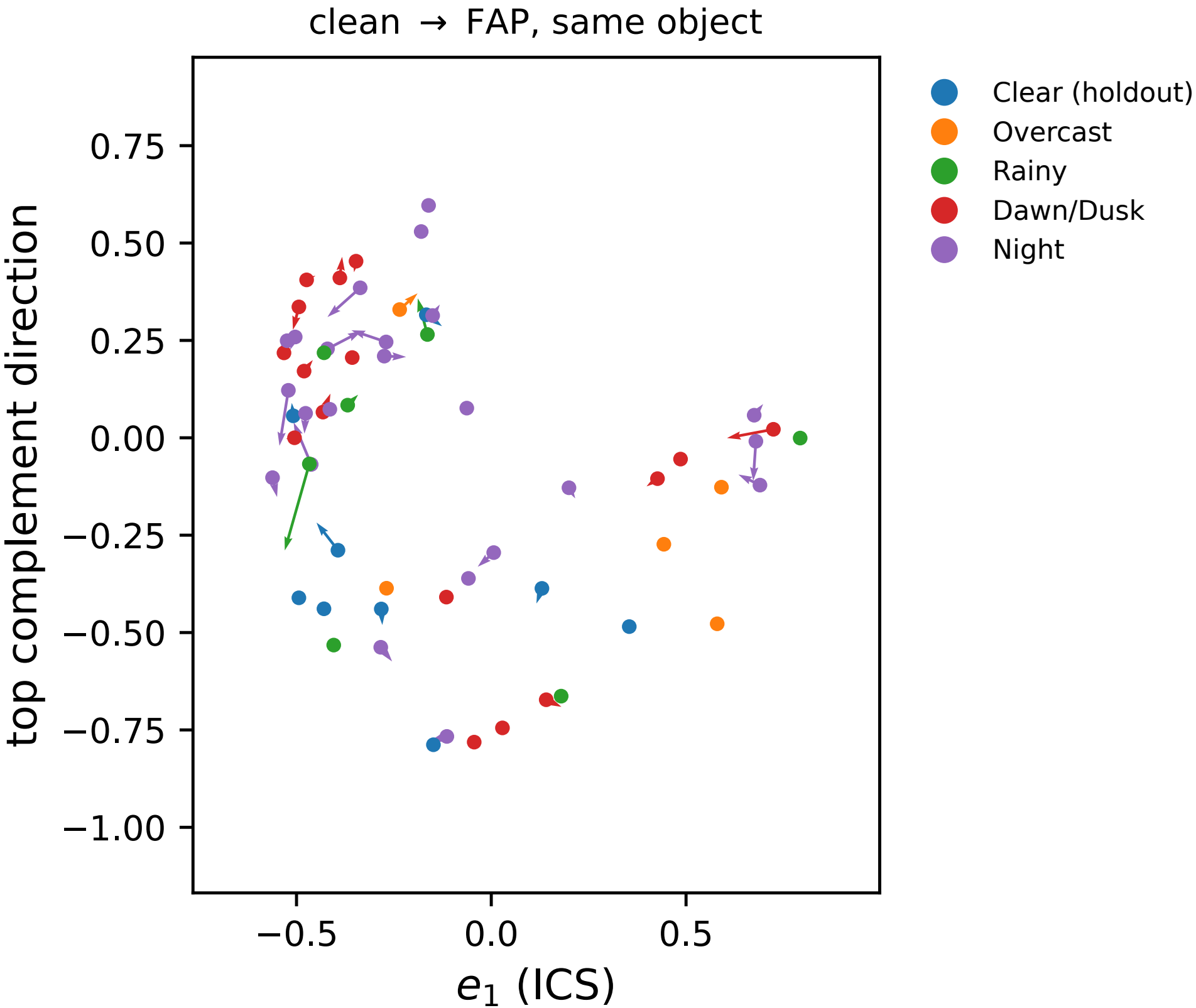
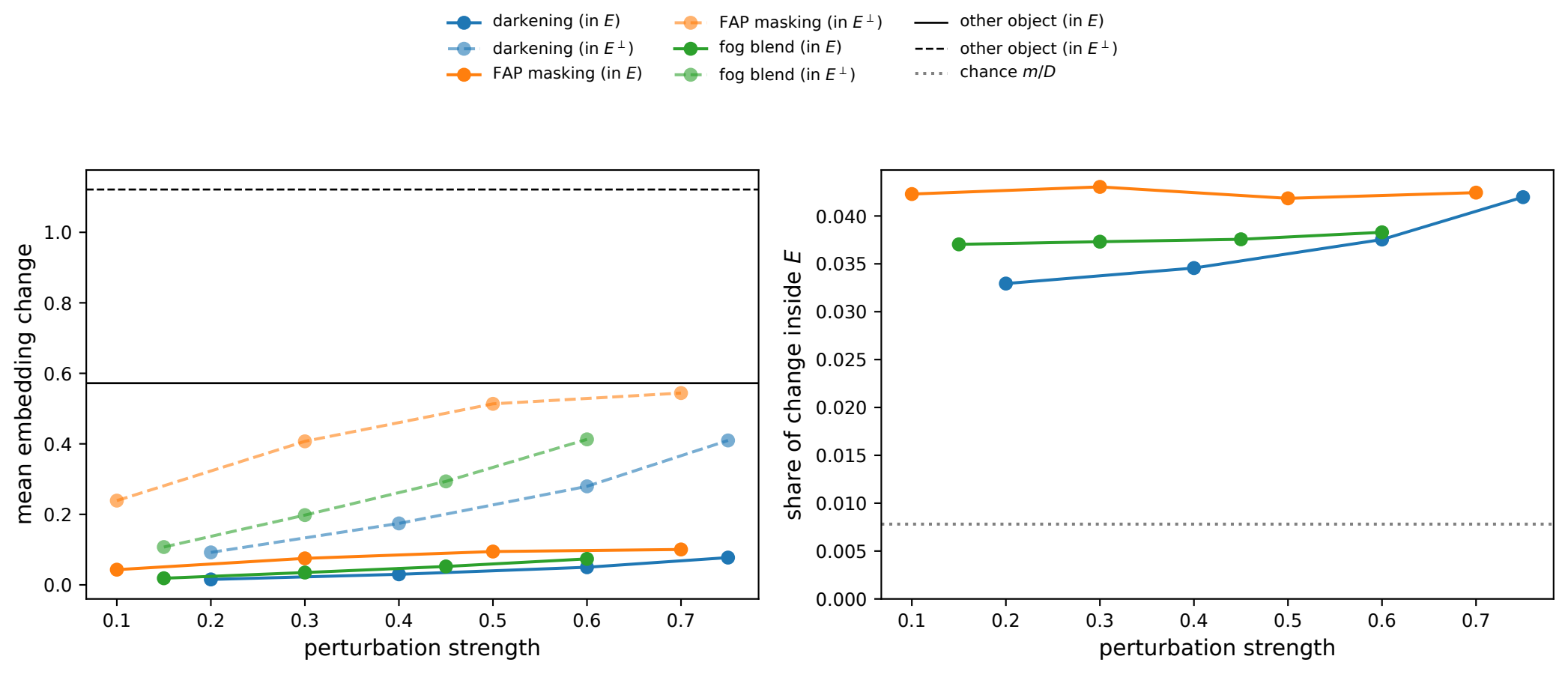
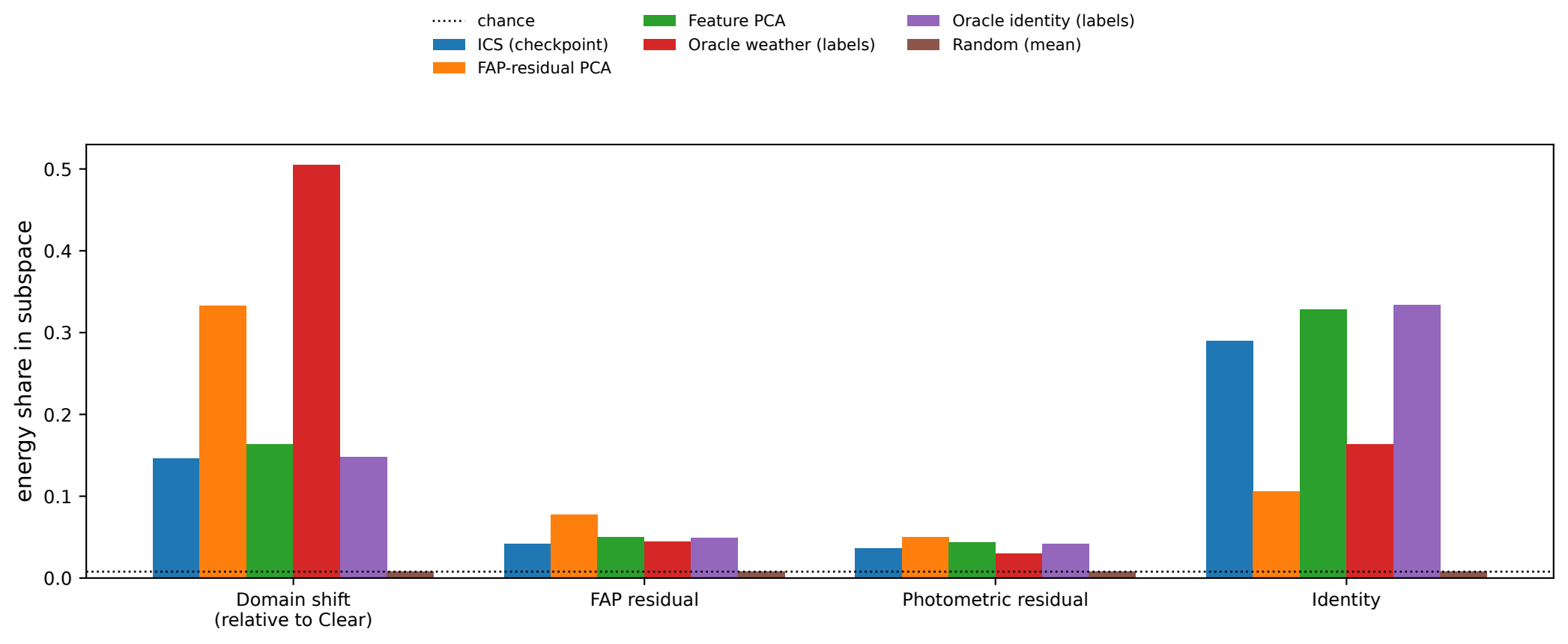

Clear (holdout)
Overcast
Rainy
Dawn/Dusk
Night
ICS subspace ($e_1, e_2$), colored by weather
Complement (t-SNE), colored by weather
ICS subspace ($e_1, e_2$), colored by identity
Complement (t-SNE), colored by identity
Highlighted tracks (video + instance)
ID 1
ID 2
ID 3
ID 4
ID 5
ID 6
ID 7
ID 8
ID 9
ID 10

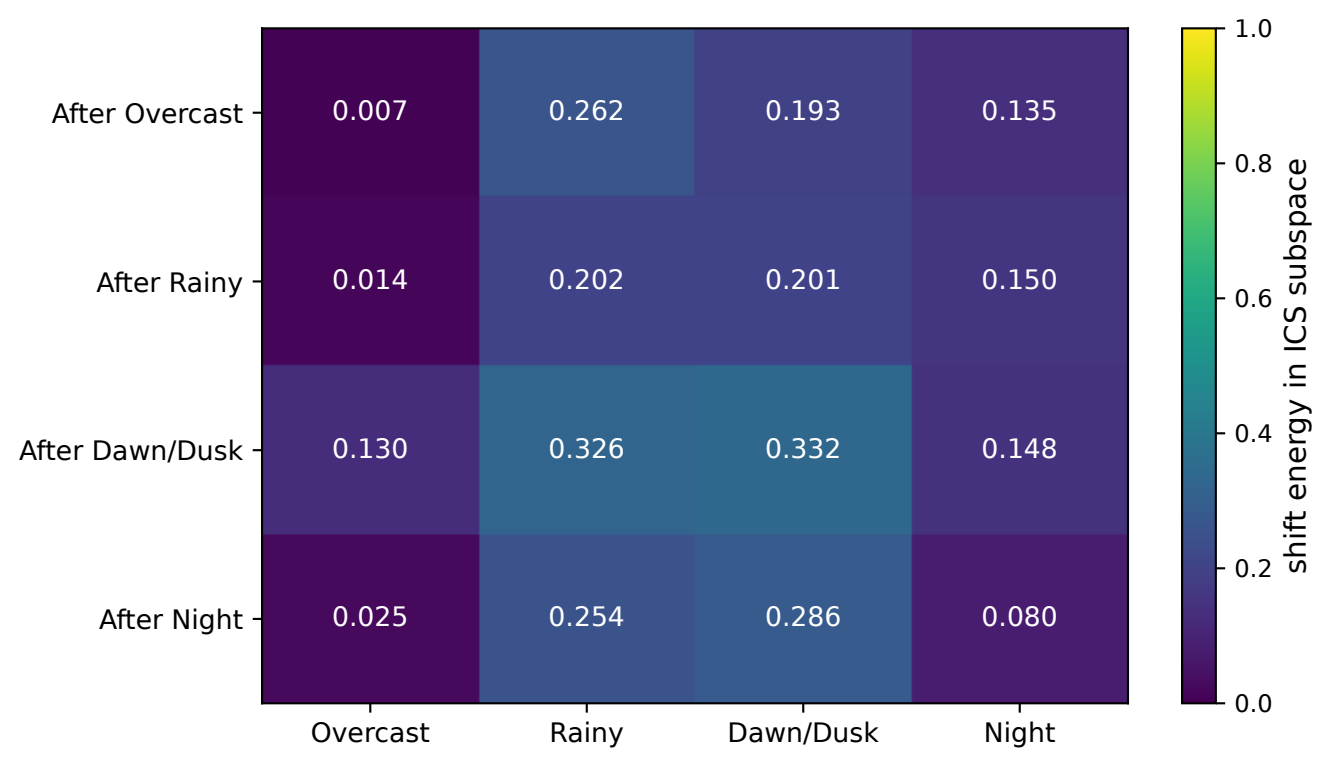

clean → FAP, same object
top complement direction
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−0.5
0.0
0.5
$e_1$ (ICS)
Clear (holdout)
Overcast
Rainy
Dawn/Dusk
Night


low $e_1$
Rainy
-0.738
Clear
-0.729
Dawn/Dusk
-0.728
Night
-0.726
Night
-0.722
Night
-0.721
Dawn/Dusk
-0.721
Overcast
-0.720
Night
-0.713
Rainy
-0.708
high $e_1$
Dawn/Dusk
+0.834
Night
+0.831
Clear
+0.828
Dawn/Dusk
+0.816
Dawn/Dusk
+0.810
Night
+0.810
Dawn/Dusk
+0.805
Dawn/Dusk
+0.803
Rainy
+0.803
Dawn/Dusk
+0.802


low $e_2$
Night
-0.766
Night
-0.759
Night
-0.751
Dawn/Dusk
-0.749
Night
-0.745
Dawn/Dusk
-0.740
Rainy
-0.737
Clear
-0.736
Clear
-0.733
Dawn/Dusk
-0.724
high $e_2$
Night
+0.817
Night
+0.781
Dawn/Dusk
+0.768
Dawn/Dusk
+0.761
Dawn/Dusk
+0.750
Night
+0.749
Rainy
+0.743
Clear
+0.741
Night
+0.741
Night
+0.739


darkening (in $E$)
darkening (in $E^\perp$)
FAP masking (in $E$)
FAP masking (in $E^\perp$)
fog blend (in $E$)
fog blend (in $E^\perp$)
other object (in $E$)
other object (in $E^\perp$)
chance $m/D$
mean embedding change
perturbation strength
share of change inside $E$
perturbation strength


chance
ICS (checkpoint)
FAP-residual PCA
Feature PCA
Oracle weather (labels)
Oracle identity (labels)
Random (mean)
energy share in subspace
0.0
0.1
0.2
0.3
0.4
0.5
Domain shift
(relative to Clear)
FAP residual
Photometric residual
Identity


After Overcast
After Rainy
After Dawn/Dusk
After Night
0.007
0.262
0.193
0.135
0.014
0.202
0.201
0.150
0.130
0.326
0.332
0.148
0.025
0.254
0.286
0.080
Overcast
Rainy
Dawn/Dusk
Night
shift energy in ICS subspace
1.0
0.8
0.6
0.4
0.2
0.0